



1. Introduction

Brain metastases (BM) remain a leading cause of morbidity and mortality in breast cancer. Although 10–20 % of women with breast cancer develop BM during their disease course, survival after CNS involvement varies markedly by subtype—from as few as 8 months in triple-negative breast cancer (TNBC) to over 60 months in HER2-positive disease. Early CNS spread is most common in TNBC and HER2-positive tumors, whereas hormone receptor-positive/HER2-negative (HR+/HER2-) cancers metastasize to the brain more gradually. That is because these genes give tumor deposits an increased likelihood of establishing a detectable or radiologically

Colonization of the brain requires tumor cells to complete a sequence of biological bottlenecks: intravasation into the bloodstream, survival in circulation, extravasation across the blood–brain barrier, and adaptation to the neural microenvironment. Genomic alterations used in the that lend tumor cells while other clinical factors—such as tumor size and nodal burden—may dominate risk at later time points.

To date, mechanistic insights into breast cancer BM rely on small cohorts or preclinical models implicating genes such as TP53, ERBB2, ST6GALNAC5, and MMP-1 in blood–brain barrier transmigration and CNS colonization. Large, multi-institutional studies that integrate detailed clinical annotation with high-throughput sequencing are needed to define robust biomarkers of brain-tropic behavior and to develop predictive tools for time to BM.

In this study, we analyze 1,019 patients from the AACR Project GENIE Biopharma Collaborative (GENIE BPC) Breast Cancer v1.2 cohort (2013–2018) with next-generation sequencing and ≥ 2 years of follow-up. We compare clinical and genomic features between patients with and without BM, estimate cumulative incidence of CNS spread (accounting for competing risks), and evaluate overall survival via Kaplan–Meier analysis. Finally, we construct a predictive model of time to brain metastasis using receptor subtype and key genomic alterations. Our findings will identify a genomic signature of CNS tropism, inform risk stratification, and guide targeted surveillance and therapeutic strategies.

2. Materials and Methods

Study Design and Cohort Assembly

We conducted a retrospective cohort study using the AACR Project GENIE Biopharma Collaborative (GENIE BPC) Breast Cancer v1.2 dataset. Patients were eligible if they:

- Had stage I–IV breast cancer
- Underwent next-generation sequencing (NGS) at DFCI, MSKCC, or VICC between 2013–2018
- Were aged 18–56 years at sequencing
- Had ≥ 2 years of potential follow-up
- Carried one of the specified OncoTree breast cancer diagnoses

A PRISMA-style flowchart (Figure 1) details inclusion/exclusion.

Data Collection and Variable Definitions

Clinical variables (age, sex, smoking status, stage, receptor subtype, insurance status) were extracted via the PRISMM™ model. Receptor subtype was categorized as Luminal A, Luminal B, HER2-positive, or Triple-Negative. Genomic data were harmonized across panels; a gene was scored “altered” if any mutation, amplification, deletion, or fusion was present.

Dates of diagnosis, sequencing, last follow-up, and brain metastasis detection were parsed in R 4.2.1.

- Overall survival (OS): time from sequencing to death or last follow-up; left truncation at sequencing
- Time to brain metastasis (TTBM): time from diagnosis to brain mets; censored at last follow-up or death if no BM

Missing values in clinical covariates (<5%) were handled by complete-case analysis.

Descriptive Statistics

Baseline characteristics are summarized in Table 1 as median (IQR) for continuous variables and n (%) for categorical variables. Comparisons used Wilcoxon rank-sum tests and Pearson’s chi-square tests with Benjamini–Hochberg correction.

Risk Modeling of Brain Metastasis and Overall Survival

- Cumulative incidence of BM was estimated via Fine–Gray subdistribution hazards models (R ``cmprsk``), treating death without BM as a competing event.
- Overall survival was assessed with Kaplan–Meier curves and log-rank tests, and multivariable Cox proportional hazards models (R ``survival``), adjusting for the same covariates.

Predictive Modeling of Time to Brain Metastasis

To create individualized risk estimates, we fit a random survival forest (RSF) using Python's ``scikit-survival``:

1. Feature engineering: One-hot encode categorical covariates; select genes with $q < 0.05$ in subdistribution-hazard models
2. Model specification: RSF with 200 trees, min node size = 5, ``max_features="sqrt"``
3. Training and tuning: 5-fold cross-validation over hyperparameters
4. Evaluation: Concordance index, time-dependent AUC, integrated Brier score
5. Interpretation: Variable importance rankings and partial dependence plots

All analyses were executed in R 4.2.1 and Python 3.9. Code and de-identified data are available in the Supplementary Materials.

3. Analysis Outline (Quick Reference)

1. Load data from Excel
2. Preprocess: factor conversion, date parsing, time-to-event variables
3. Descriptive stats: Table 1, Figure 1
4. Risk modeling:
 - Fine–Gray for brain mets
 - Cox PH for overall survival
5. Feature selection: significant genes from models
6. Predictive modeling:
 - One-hot encode

- Fit RSF
- Evaluate performance

7. Interpretation: importance + partial dependence

8. Reproducibility: R + Python scripts

4. Teaching Notes

One-Hot Encoding

Transforms categorical variables into binary columns—one per category. Prevents models from assuming ordinal relationships and enables inclusion in tree-based or linear models.

Fine-Gray Regression

Used when death competes with the event of interest (e.g., brain mets). Models subdistribution hazard to estimate cumulative incidence while accounting for competing risks.

Random Forests for Nonlinear Data

Tree-based models capture complex interactions and nonlinear effects without needing to specify them. Ensemble averaging improves generalization. Use residual plots, GAMs, or partial dependence plots to demonstrate nonlinearity.

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